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Best of beaches

Nine Brevard County residents gathered last month to rate sand samples from Florida beaches. They commented on the sands' color and texture and gave their opinion about which they considered the "best." Each sand got an overall ranking and ranks for color and texture. Here are the results:



1. ST. JOSEPH'S BEACH
Gulf County
COLOR: 1 TEXTURE: 1
DESCRIPTION: White, powdery soft.



2. FORT MYERS BEACH
Lee County
COLOR: 3 TEXTURE: 2
DESCRIPTION: Almond tan, soft.



3. DESTIN
Okaloosa County
COLOR: 2 TEXTURE: 3
DESCRIPTION: Egg-shell white, sugary soft.



4. PENSACOLA BEACH
Escambia County
COLOR: 4 TEXTURE: 4
DESCRIPTION: White, sugary.



5. JACKSONVILLE BEACH
Duval County
COLOR: 8 TEXTURE: 5
DESCRIPTION: Light gray, salty texture.



6. DAYTONA BEACH
Volusia County
COLOR: 7 TEXTURE: 6
DESCRIPTION: Tan with orange specks, soft and smooth.



7. PANAMA CITY BEACH
Bay County
COLOR: 6 TEXTURE: 8
DESCRIPTION: Yellow-tan, grainy.



8. ANNA MARIA ISLAND
Manatee County
COLOR: 5 TEXTURE: 13
DESCRIPTION: White with black and gray specks, coarse with shattered shells.



9. LIDO BEACH
Sarasota County
COLOR: 10 TEXTURE: 9
DESCRIPTION: Dark gray with black accents, smooth and grainy.



10. FLAGLER BEACH
Flagler County
COLOR: 12 TEXTURE: 7
DESCRIPTION: Orange, smooth, but prickly with tiny shell fragments.



Hands-on panel sifts state's sand



Photos by Steve Dozier, FLORIDA TODAY

Sampling sand: Michael Malone, 7, of Merritt Island in Brevard County samples sand from Jacksonville Beach in Duval County. Michael's favorite was the sand from Fort Myers Beach in Lee County. Top: Michael's dad, Giles Malone, sinks his toes into one of the softer samples.

Coast's grainy carpet has wide range of color, feel

By Chris Kridler

You could make a cake out of some of Florida's sand if texture and color reflected reality: Flour and sugar. Bright and white.

But you wouldn't want to eat it if all of the state's coarse and colorful grains were mixed in, given the findings of some brave sand-sifters who dipped their fingers and toes in some of Florida's finest.

Facing plastic pans filled with samples from 19 beaches, nine judges, including a family of six, set about rating the mixtures of ground rock, coral and shells that beach towns depend on to lure tourists — and to keep their condos from sliding into the ocean.

They all kind of looked alike at first.

"I don't know, it just kind of looks like sand, nothing extraordinary," said Emily Bourne, 18, of Palm Bay, as she checked out brown grains from St. Johns County, the home of St. Augustine. "I don't know about the texture. I don't think I want to lie on it, because it has little shells and stuff in it."

The sands ranged in color from orange to ash to snow. The judges scored them on color and texture without knowing their origins.

The kids were, shall we say, very hands-on. Michael Malone, 7, was up to his elbows in the lovely white sand from Fort Myers Beach, which finished second. He looked as if he were kneading bread. "It's good. I like it," he said.

He even liked the sand with the tiny spider in it, from Pensacola. "I like spiders, and I like this type of sand. It feels good," he said. His sister Megan, 11, agreed, saying it was soft and pretty.

There were many raves for the Pensacola sand, which ranked fourth, after Destin. Pensacola's wasn't the finest grain, said Abigail Lee, 17, of Rockledge, but it was her favorite.

"This one feels really natural and looks most natural," she said. "It's not like an artificial light, and it feels good on the toes."

Too fine to be gravel, too coarse to be dust. That's sand. In some parts of the world, there is singing sand, which hums with the wind. There's magnetic sand, which is high in iron. There's high-silica sand, the kind used in glass.

And then there's the dredged kind. The sample from Lido Key, near Sarasota, was born of a renourishment project that darkened the "powder white" sand local hotels like to advertise.

It's the color of cinders and provoked intense reactions. It ranked in the middle of the samples.

Giles Malone, president of the Cocoa Beach Pier and patriarch of the Merritt Island family digging through the basins, called it "very unusual" despite a fine texture.

"I couldn't imagine where it would be from. It's almost like it has some volcanic rock in it or something, it's so dark."

His daughter Victoria, 8, said it's "all seedy," but Parth Modi, 16, of Titusville, liked it best.

"It looks pretty cool," he said, admiring the even size of the granules. He had one shoe off so he could dig into all the samples with the same set of toes, maintaining "scientific integrity," he said.

He also was intrigued by the orange sand from Flagler County, which might as well be from Mars. "A whole landscape of this kind of sand would look kind of strange," he said.

Little Grace Malone, 2, would say only that it was "pink."

Sand collectors call themselves arenophiles. Sand critics aren't so refined. "Ow, ow, ow, ow!" Michael said of the light gray and white, coarse handfuls from the west coast's Sanibel Island. "Those shells are pokey!"

His reaction may explain why Sanibel is known more for its shells than its sand — and came in last place. Michael's mother, Alison Malone, called it "gritty and grainy and just crunchy."

Even when Floridians aren't on the beach, they're surrounded by sand. It's used in concrete, so it's part of the condos and patios built by the tide of people overwhelming the state's eroding beaches.

Based on the scores, the panel would rather experience the beach in the Panhandle's St. Joseph Peninsula State Park than anywhere else. Its sand was ranked best overall, and it rated tops for color and texture.

The judges are in good company. Stephen Leatherman, aka "Dr. Beach," ranked St. Joseph No. 1 in his 2002 list of the best U.S. beaches.

The St. Joseph sand is comfortable, said Giles Malone. "You could wiggle your toes in that one all afternoon without any problem at all."

Grace, the tot, was trying out each basin by stuffing a beach ball into it. When she got to the St. Joseph sand, she wordlessly left the ball, pointed to it and walked away.

Enough said.

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Picking the best: Abigail Lee, 17, of Rockledge liked the sugary, white sand of Pensacola Beach in Escambia County. "It feels good on the toes," Lee said.

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Andrew West, The News-Press

Sandy springboard: Abdul Serroukh on vacation from the Ringling Bros. and Barnum & Bailey Circus practices his flips on Lido Beach in Sarasota. More than 21 miles of beach are deemed critically eroded along Sarasota County's coast. Lido Beach was renourished from 30 feet to 200 feet in May 2001 in a $4.18 million project, however, some county residents say they don't like the sand. So Sarasota and Longboat Key are paying $400,000 to blanket the existing beach with 110,000 cubic yards of white sugar sand dredged from New Pass.

Perfect sand a rare find

Search for right stuff difficult as underwater deposits decline

By Anton Caputo

Lifelong Panama City resident Rick LaMarre has mixed feelings when he scans the beach that was all but nonexistent three years ago.

"I'm sure glad the beach is back, but I wish it didn't have to come at the expense of all this trash," he said. "It used to be like sugar, so white and bright that it hurt your eyes to look at."

In its place is sand ranging from off-white to gray, tan and brown, the result of a major renourishment in which sand was pumped from the nearby Gulf of Mexico floor. Where once were only the pure granules of crystal quartz is sand mixed with shells and rocks.

It's not exactly trash, but it's not the powdery, white sand Panhandle residents are used to, and that has some nervous.

Shift the scene 550 miles downstate and many South Florida residents would be happy to have Panama City's tan-gray sand. They'd be happy to have any sand. After about three decades of aggressive beach renourishment, pumping sands from deposits up to 20 miles away, they've depleted the vast stores that settled between the coral reefs off Miami-Dade County's shores.

Sand's a hot commodity in the Sunshine State.

It's hard to find, getting more expensive and pushing some counties to extremes in the quest for the surprisingly rare substance.

"Sand is valuable," said Stephen "Dr. Beach" Leatherman, director of the Florida International University's Laboratory for Coastal Research. "It's like gold down here."

Panhandle counties are spending millions of dollars searching for deposits of pure, white sand to replenish their beaches. Search boats use technology similar to sonar to map the near-shore ocean floor, looking for likely sand deposits. Once found, crews will take hundreds of 20-foot-long, 3-inch diameter core samples and analyze them for color and grain size.

After two years and more than $800,000 of study, Navarre Beach is still looking for sand palatable to local officials and residents.

South Florida counties are spending millions more looking for sand at all.

Last year, Broward County contracted the services of an Australian defense firm to help map the ocean floor via a high-tech airplane and aerial laser. The technique gave local officials a better idea of just how much near-shore sand can be dredged without harming the environmentally sensitive coral reefs.

Now, state officials are considering using the method statewide.

"It gives us a view of the ocean floor," said Stephen Higgins, the county's beach erosion administrator. "We can see ledges, reefs and old (sand) borrow sites."

Higgins figures he has enough sand left for one more big renourishment project that is scheduled to start this fall. The $47 million job will rebuild almost 12 miles of beaches in the southern two-thirds of the county. About $10 million of that went for engineering and sand search.

Strict standards

"After that we'll have to look elsewhere," he said. "We're probably going to look at doing frequent smaller infusions of sand in hot spots by truck. But you don't want to do a big project that way. You could probably run trucks for the rest of your life and not build up enough beach to notice."

Those trucks get sand from quarries around the state that usually market to construction companies as fill dirt. Occasionally, they produce sand that meets the state's regulation for beach use. These include standards for grain size and clay and gravel content. The Miami area has already run out.

Because of that, the U.S. Army Corps of Engineers is requiring any company that wants to bid on Miami-Dade's next beach renourishment project to supply its own sand.

From a beach engineering standpoint, most any sand will do. The larger and heavier the grains of sand, the more the beach resists eroding waves.

Conversely, larger quantities are needed when finer sand is used because it is more easily washed away. But from an environmental standpoint, the wrong sand can harm indigenous wildlife, especially nesting sea turtles.

From a political and public relations standpoint, the wrong sand can be a disaster. On the Gulf coast, residents feel connected to their white, powdery sand and communities use it to lure tourists.

People already complain about the color in Panama City and Lido Beach near Sarasota. In both areas, dredging crews are replacing darker sand with lighter sand to allay residents' complaints.

In September, crews on Lido Beach will push the top 2 feet of sand that was put there last year into the nearby surf. It will be replaced with lighter sand dredged from the nearby pass as part of a Corps of Engineers project. The touch-up work will cost the city $400,000.

"People were extremely happy that we did the renourishment, but they grumbled about the color," Sarasota City Engineer Dennis Daughters said. "The sand we dredged from offshore was kind of gray, but if you asked some residents, they would probably say it was black."

Matching the sand takes a lot of work and money, but the technology to match the color is remarkably simple, said coastal engineer Erik Olsen of Olsen and Associates in Jacksonville.

"It's like going to a paint store," he said.

It is done with the aid of a color sample rated by the Munsell Color Chart, which rates color on a scale of 1 to 10, with 10 being pure white. The darkest of the tan sand used on Panama City Beach, which was dredged from a couple hundred yards to 1.6 miles offshore, rates about a 5 or 6 on the scale for lightness. The areas being lightened will rank about a 7. Typical Panhandle sand rates about 9.

On Pensacola Beach, where Olsen is helping plan a similar project, engineers are shooting for an average color rating of 9.25.



Buy Bahamian?

With sand supplies running short, people are looking far and wide for new sources. One potential source for South Florida viewed as virtually limitless are the Bahama Banks, which contain billions of tons of light-tan aragonite sand.

The sand is coarser than natural southeastern Florida sand, which means it better resists erosion. It is also uniform in shape and contains little or no silt that can discolor beaches. Island miners now market the sand for agricultural use and use in Portland cement.

It's been used on an experimental renourishment job on South Florida's swanky Fisher Island near Miami, but the Bahamas' Ministry of Tourism recently balked at the notion of making the sand available to renourish Florida beaches.

To make matters more difficult, U.S. dredging companies have thrown their political weight behind "buy American" legislation that blocks sand importation.

Despite all the hurdles, Olsen, who participated in the Fisher Island project, thinks economics will eventually open the way for the Bahamian sand.

As is, the $10 a cubic yard it is estimated to cost is competitive with the going rate in sand-starved Miami-Dade. Even if the Bahamas continues to balk at the notion of selling sand to its tourism competition, other Caribbean Islands are showing signs of being more than willing to enter the market.

Officials with the Turks and Caicos Islands have attended conferences with plastic bags of sand to show off.

"I actually found myself walking through the Miami Airport one day with a little baggie of white powder," said Phil Flood, project manager with the Florida Department of Environmental Protection's Offices of Beaches and Coastal Systems.

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18. SANIBEL ISLAND
Lee County
COLOR: 18 TEXTURE: 18
DESCRIPTION: Whitish gray, coarse and crunchy with lots of large, broken shell fragments.



17. PLAYALINDA BEACH
Brevard County
COLOR: 17 TEXTURE: 17
DESCRIPTION: Brownish-orange and grainy with tiny crunched shells.



16. ST. AUGUSTINE
St. Johns County
COLOR: 15 TEXTURE: 15
DESCRIPTION: Tannish-orange, coarse.



15. MIAMI BEACH
Miami-Dade County
COLOR: 13 TEXTURE: 16
DESCRIPTION: Gray with tan and white specs, grainy.



11. VERO BEACH
Indian River County
COLOR: 11 TEXTURE: 10
DESCRIPTION: Whitish-orange, smooth but gritty.



12. ST. PETE BEACH
Pinellas County
COLOR: 9 TEXTURE: 12
DESCRIPTION: White, soft, but mixed with finely ground shell.



13. FORT LAUDERDALE
Broward County
COLOR: 16 TEXTURE: 11
DESCRIPTION: Brown-orange, smooth but grainy.



14. PALM BEACH
Palm Beach County
COLOR: 14 TEXTURE: 14
DESCRIPTION: Grayish-orange with black specks, gritty.